



# **BHABHA UNIVERSITY, BHOPAL**

## **Scheme of Examination**

### **Seventh Semester-Bachelor of Technology (Electrical Engineering) Session 2020-21**

#### **SPECIALIZATION – Electrical Engineering**

| S.No. | Subject Code | Subject Name              | Periods per week |           |           | Credits   | Maximum Marks (Theory Slot) |               |                    | Maximum Marks (Practical Slot) |                                                   | Total Marks |
|-------|--------------|---------------------------|------------------|-----------|-----------|-----------|-----------------------------|---------------|--------------------|--------------------------------|---------------------------------------------------|-------------|
|       |              |                           | L                | T         | P         |           | End Sem. Exam               | Mid sem Tests | Assignments / Quiz | End Sem. Practical / Viva      | Practical Record/ assignment/ Quiz / Presentation |             |
| 1     | EE-71        | Electrical machine Design | 3                | 1         | 2         | 5         | 70                          | 20            | 10                 | 30                             | 20                                                | 150         |
| 2     | EE-72        | Elective-I                | 4                | 1         |           | 5         | 70                          | 20            | 10                 |                                |                                                   | 100         |
| 3     | EE-73        | Elective-II               | 4                | 1         |           | 5         | 70                          | 20            | 10                 |                                |                                                   | 100         |
| 4     | EE-74        | Electrical Cad Lab        | -                | -         | 4         | 2         | -                           | -             | -                  | 50                             | 50                                                | 100         |
| 7     | EE-75        | Major Project-I           | -                | -         | 8         | 4         | -                           | -             | -                  | 100                            | 50                                                | 150         |
| 8     | EE-76        | Industrial Training       | -                | -         | 6         | 3         | -                           | -             | -                  | 100                            | 50                                                | 150         |
|       |              | <b>Total</b>              | <b>11</b>        | <b>03</b> | <b>20</b> | <b>24</b> | <b>210</b>                  | <b>60</b>     | <b>30</b>          | <b>280</b>                     | <b>170</b>                                        | <b>750</b>  |

| <b>Elective-I</b>                                 |  | <b>Elective-II</b>                                     |  |
|---------------------------------------------------|--|--------------------------------------------------------|--|
| EE-72(A) Electrical Drive                         |  | EE-73(A) Power Electronics Application to Power System |  |
| EE-72(B) HVDC & FACTS                             |  | EE-73(B) Advance Control System                        |  |
| EE-72(C) Generalized Theory of Electrical Machine |  | EE-73(C) Soft Computing Technique and Application      |  |

**Theory Credits : 12**  
**Practical Credits: 12**  
**Total Credits : 24**

**Theory Marks : 210**  
**Practical Marks: 280**  
**Quiz, Mid Term, Lab. Work: 260**

**Total: 750**



**BHABHA UNIVERSITY, BHOPAL**

**COURSE CURRICULUM  
FACULTY OF ENGINEERING**

**Undergraduate Programme**

**(Course effective from Academic Year 2020-21)**

**SYLLABUS OF COURSES TO BE OFFERED**

**“Bachelor of Technology”**

**Disclaimer:** The Undergraduate programme in Bachelor of Technology syllabus & schemes is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

**Dean Faculty of Engineering, Bhabha University**

**NH-12, Jatkhedi, Hoshangabad Road, Bhopal (M.P.) – 462026**

visit us at: [www.bhabhauniversity.edu.in](http://www.bhabhauniversity.edu.in), Mail: [info@bhabhauniversity.edu.in](mailto:info@bhabhauniversity.edu.in)



**Bachelor of Technology VII  
Semester  
Electrical Engineering (EE)**

**NH-12, Jatkhedi, Hoshangabad Road, Bhopal (M.P.) – 462026**  
Visit us at: [www.bhabhauniversity.edu.in](http://www.bhabhauniversity.edu.in), Mail: [info@bhabhauniversity.edu](mailto:info@bhabhauniversity.edu).

# **Bachelor of Technology**

## **Branch- Electrical Engineering**

### **EE-71 Electrical Machine Design**

#### **Unit-I**

Fundamental Aspects of Electrical Machine Design: Design of Machines, Design Factors, Limitations in design, Modern Trends in design, manufacturing Techniques. Electrical Engineering Materials: Desirabilities of Conducting Materials, Comparison of Aluminium and Copper wires. Ferromagnetic Materials: Soft Magnetic materials – Solid Core Materials, Electrical Sheet and Strip, Cold Rolled Grain Oriented Steel. Insulating Materials: Desirable Properties, Temperature Rise and Insulating Materials, Classification of Insulating materials based on Thermal Consideration.

#### **Unit-II**

Design of DC Machines: Output Equation, Choice of Specific Loadings and Choice of Number of Poles, Main Dimensions of armature, Design of Armature Slot Dimensions, Commutator and Brushes. Estimation of Ampere Turns for the Magnetic Circuit. Dimensions of Yoke, Main Pole and Air Gap. Design of Shunt and Series Field Windings.

#### **Unit-III**

Design of Transformers: Output Equations of Single Phase and Three Phase Transformers, Choice of Specific Loadings, Expression for Volts/Turn, Determination of Main Dimensions of the Core, Estimation of Number of Turns and Conductor Cross Sectional area of Primary and Secondary Windings, No Load Current. Expression for the Leakage Reactance of core type transformer with concentric coils, and calculation of Voltage Regulation. Design of Tank and Cooling (Round and Rectangular) Tubes.

#### **Unit-IV**

Design of Three Phase Induction Motors: Output Equation, Choice of Specific Loadings, Main Dimensions of Stator. Design of stator slots and Winding, Choice of Length Air Gap, Estimation of Number of Slots for Squirrel Cage Rotor. Design of Rotor Bars and End Ring. Design of Slip Ring rotor. Estimation of No Load Current and Leakage Reactance.

#### **Unit-V**

Design of Three Phase Synchronous Machines and Computer aided Design (CAD): Design of Three Phase Synchronous Machines: Output Equation, Choice of Specific Loadings, Short Circuit Ratio, Main Dimensions of Stator. Design of stator slots and Winding. Design of Salient and non-salient Pole Rotors. Magnetic Circuit and Field Winding.

# **Bachelor of Technology**

## **Branch- Electrical Engineering**

### **REFERENCE BOOKS:-**

- 1 Sawhney, A.K., 'A Course in Electrical Machine Design', Dhanpat Rai & Sons, New Delhi, 1984
- 2 M.V. Deshpande "Design and Testing of Electrical Machine Design" Wheeler Publications, 2010.
- 3 A. Shanmuga Sundaram, G. Gangadharan, R. Palani 'Electrical Machine Design Data Book', New Age International Pvt. Ltd., Reprint, 2007
- 4 R.K. Agarwal "Principles of Electrical Machine Design" Esskay Publications, Delhi, 2002

### **EXPERIMENT LIST:-**

- To study the laboratory measuring instruments and their application.
- To Understand the Nameplate Data/specification of Electrical Machines.
- To demonstrate construction and working of DC Machines with their Functional Details.
- To demonstrate the constructional part and working principle of AC Motors.
- To demonstrate the constructional part and working principle of Transformers.
- To perform tests for performance calculation of 1-phase and 3-phase Transformer.
- To demonstrate construction and working of Magnetic Contactor.
- To study the Earth leakage circuit breaker (ELCB).
- To demonstrate the working principle of miniature circuit breaker (MCB).
- To demonstrate constructional parts and working principle of induction type energy meter (KWh meter).
- To demonstrate the construction and working principle of thermal overload relay.
- To study Transformer oil insulation test.
- To demonstrate the working principle of Earth tester.
- Study, Industrial applications of machines and drives for desired performance.

# **Bachelor of Technology**

## **Branch-Electrical Engineering**

### **ELECTIVE-I**

#### **EE-72(A) Electrical Drives**

##### **Unit I**

Basic Concepts of Electric Drives Elements of drive systems, Requirement of electric drives, Rating & Selection of drives, groups and individual drives, Constant power and Constant torque drives. Motor Mechanism dynamics Review of Characteristics of AC & DC motors, load characteristic, load-drive speed torque characteristics, quadrant speed torque characteristics. Mechanical Systems Stability of Electric drives, referred moment of inertia and torque of motor load combination, load equalization.

##### **Unit II**

DC Drives Starting & Braking of conventional, Phase controlled and chopper controlled drives, Transient & Steady state analysis, Energy recovery systems.

##### **Unit III**

Induction Motor Drives Conventional method of Starting braking and speed control, PWM, (VSI) Voltage source Inverter and Current Sources (CSI) fed IM drives, cycloconverter fed drive, Vector control drives. Slip Controlled IM Drives Review of Conventional methods & converter controlled - Cramers & Scherbius drives; rotor impedance control.

##### **Unit IV**

Synchronous Motors Drives VSI and CSI fed; self-controlled-Brushless & commutatorless dc & ac motor drives.

##### **Unit V**

Special Drives: Fundamentals of Switched reluctance motors, Stepper Motors, Permanent Magnet Motor Introduction to vector control; Digital control of drives. Case Studies Electric traction, steel & cements plants, textile & paper mills, machine tool drive and CNC, electric cars.

#### **REFERENCE BOOKS:-**

- Pillai S.K. "A first course on Electrical Drives", Second edition, Wiley Eastern.
- Ned Mohan Electrical Machine Drive WILEY INDIA
- Dubey G.K., "Power Semiconductor Controlled Drives", PHI,
- Dubey G.K., "Fundamentals of Electrical Drives". Narosa Publishing House.
- Bose B.K., "Power Electronics and AC Drives", PHI Learning.
- Murphy M.D., and Tumbuli F., "Power Electronic Control of AC Motors", Pergamon Press, Oxford
- University Press. P.V. Rao, "Power semiconductor Drives", BS Publications
- S. Shiva Nagaraju power semiconductor drive PHI learning

**Bachelor of Technology**  
**Branch- Electrical Engineering**

# **Bachelor of Technology**

## **Branch-Electrical Engineering**

### **EE-72(B) HVDC & FACTS**

#### **UNIT I**

Basic Issues Involved in Bulk Power Transmission, Review of basics of power transmission networks-control of power flow in AC transmission line- Analysis of uncompensated AC Transmission line- Passive reactive power compensation, Principle of Transmission system compensation, Need for FACTS controllers- types of FACTS controllers and Benefits.

#### **UNIT II**

STATIC VAR COMPENSATOR (SVC) and Purpose Voltage control by SVC – Advantages of slope in dynamic characteristics- Influence of SVC on system voltage, Design of SVC voltage regulator, Modeling of SVC for power flow and stability studies, Applications- Enhancement of transient stability, Steady state power transfer, Enhancement of Power system damping, Prevention of voltage instability

#### **UNIT III**

THYRISTOR AND GTO THYRISTOR CONTROLLED SERIES CAPACITORS (TCSC and GCSC) Concepts of Controlled Series Compensation – Analysis of TCSC-GCSC, Different modes of operation, Modeling of TCSC and GCSC for load flow studies- modeling TCSC and GCSC for stability studies- Applications of TCSC and GCSC, SSR mitigation.

#### **UNIT IV**

VOLTAGE SOURCE CONVERTER BASED FACTS CONTROLLERS Static synchronous compensator (STATCOM)- Static synchronous series compensator (SSSC)- Operation of STATCOM and SSSC- Power flow control with STATCOM and SSSC- Modeling of STATCOM and SSSC for power flow studies – operation of Unified and Interline power flow controllers (UPFC and IPFC).

#### **UNIT V**

CONTROLLERS AND THEIR CO-ORDINATION FACTS Controller interactions – SVC–SVC interaction-co-ordination of multiple controllers using linear control techniques – Quantitative treatment of control coordination.

#### **REFERENCE BOOKS:-**

1. Mohan Mathur, R., Rajiv K. Varma, Thyristor –Based FACTS Controllers for Electrical Transmission Systems, IEEE press and John Wiley & Sons, Inc, 2002.
2. K.R. Padiyar, FACTS Controllers in Power Transmission and Distribution, New Age International (P) Ltd., Publishers, New Delhi, Reprint, 2008.
3. A.T. John, Flexible AC Transmission System, Institution of Electrical and Electronic Engineers (IEEE), 1999.
4. Narain G. Hingorani, Laszlo Gyugyi, Understanding FACTS Concepts and Technology of Flexible AC Transmission System, Standard Publishers, Delhi, 2001.
5. V.K. Sood, HVDC and FACTS controllers- Applications of Static Converters in Power System, Kluwer Academic Publishers, 2004.

# **Bachelor of Technology**

## **Branch-Electrical Engineering**

### **EE-72(C) Generalized Theory of Electrical Machine**

#### **Unit-I**

Review : Primitive machine, voltage and torque equation. Concept of transformation change of variables & m/c variables and transform variables. Application to D.C. machine for steady state and transient analysis, and equation of cross field commutator machine.

#### **Unit-II**

Induction Machine : Voltage, torque equation for steady state operation, Equivalent circuit, Dynamic performance during sudden changes in load torque and three phase fault at the machine terminals. Voltage & torque equation for steady state operation of 1- $\phi$  induction motor & synchronous motor.

#### **Unit-III**

Synchronous Machine: Transformation equations for rotating three phase windings, Voltage and power equation for salient and non salient alternator, their phasor diagrams, Simplified equations of a synchronous machine with two damper coils.

#### **Unit-IV**

Operational Impedances and Time Constants of Synchronous Machines: Park's equations in operational form, operational impedances and  $G(P)$  for a synchronous machine with four Rotor Windings, Standard synchronous machine Reactances, time constants, Derived synchronous machine time constants, parameters from short circuit characteristics.

#### **Unit-V**

Approximate Methods for Generator & System Analysis: The problem of power system analysis, Equivalent circuit & vector diagrams for approximate calculations, Analysis of line to line short circuit, Application of approximate method to power system analysis.

#### **REFERENCE BOOKS:-**

- P.C.Krause, Analysis of Electric Machinery, Wiley India.
- B.Adkins, The General theory of Electrical Machines.
- B.Adkins & R.G.Harley, The General theory of AC Machines.
- P.S.Bhimbra, Generalised theory of Electrical m/c White & Woodson, Electro Mechanical Energy Conversion.

# **Bachelor of Technology**

## **Branch-Electrical Engineering**

### **ELECTIVE-II**

#### **EE-73(A) Power Electronics Application to Power System**

##### **UNIT-I**

Steady state and dynamic problems in AC systems: Flexible AC transmission systems (FACTS), Principles of series and shunt compensation, Description of static var compensators (SVC), Thyristor Controlled series compensators (TCSC), Static phase shifters (SPS), Static condenser (STATCON), Static synchronous series compensator (SSSC) and Unified power flow controller (UPFC).

##### **UNIT-II**

Modelling and Analysis of FACTS controllers: Control strategies to improve system stability, Power Quality problems in distribution systems.

##### **UNIT-III**

Harmonics: Harmonics creating loads, modelling, harmonic propagation, Series and parallel resonances, harmonic power flow, Mitigation of harmonics, filters, passive filters, Active filters, shunt, series hybrid filters, voltage sags & swells, voltage flicker, Mitigation of power quality problems using power electronic conditioners, IEEE standards, HVDC Converters and their characteristics, Control of the converters (C and CEA), Parallel and series operation of converters.

##### **UNIT-IV**

Active Power Controllers: Dynamic static synchronous controllers, D – STATCOM, Dynamic static synchronous series controllers, dynamic voltage restorer, AC/AC voltage regulators.

##### **UNIT-V**

Energy Storage Systems: Introduction, structure of power storage devices, pumped – storage hydroelectricity, compressed air energy storage system, flywheels, battery storage, hydrogen storage, superconducting magnet energy storage, super capacitors, applications of energy storage devices.

#### **REFERENCE BOOKS:-**

1. N.G. Hingorani & Laszlo Gyugyi, Understanding FACTS, IEEE Press, 2000.
2. E.F. Fuchs & Mohammad A.S. Masoum, Power Quality in Power Systems and Electrical Machines, Elsevier Academic Press 2008.
3. K.R. Padiyar, FACTS controllers in power transmission and distribution, New Age International publishers, New Delhi, 2007.
4. K.R. Padiyar, HVDC Power Transmission Systems, New Age International publishers, New Delhi, 1999.

# **Bachelor of Technology**

## **Branch-Electrical Engineering**

### **EE-73(B) ADVANCED CONTROL SYSTEM**

#### **UNIT-I**

Review of Linear Control System: Modelling through differential equations and difference equations, State space method of description and its solution, Discretization of continuous-time state space model, Laplace and z-domain analyses of control systems, Controllability, Observability & Stability, Bode & Nyquist analysis, Root Loci, Effect of load disturbance upon control actions.

#### **UNIT-II**

Development of feedback control law through state space technique, Modal control, Pole placement problem.

#### **UNIT-III**

Variable Structure Control and its applications. Examples on variable structure control.

#### **UNIT-IV**

Control of nonlinear dynamics: Lyapunov based control function, Phase plane technique, Lyapunov Stability analysis.

#### **UNIT-V**

Optimal Control: Calculus of variation, Euler-Lagrange equations, Boundary conditions, Transversality condition, Bolza problem, Pontryagin's maximum principle.

#### **REFERENCE BOOKS:-**

1. Automatic Control System – B.C. Kuo, PHI, New York, 1975.
2. Modern Control Engineering: K. Ogata, PHI, New Delhi, 1992.
3. Digital Control Systems – B.C. Kuo, Oxford Pub.
4. Discrete-Time Control Systems – K. Ogata, PHI, New Delhi
5. Advanced Control Systems N. S. Sarkar, PHI Learning
6. Control System Engineering S. Nise, Wiley India

# **Bachelor of Technology**

## **Branch-Electrical Engineering**

### **EE-73(C) Soft Computing Techniques & Application**

#### **UNIT-I**

Review of probability theory: Random variable, distribution functions, function of random variable. generation of random digit, and random variants from various distribution function, Monte Carlo simulation, sampling distributions station evolution using MCS, confidence interval, coefficient of variation.

#### **UNIT-II**

Evolution of ANN, Artificial neurons activation functions general network structure  $\delta$ -rule, and backpropagation rule of training, RBF and FLN network.

#### **UNIT-III**

Drawback of classical optimization techniques, genetic algorithm; binary and real parameter GA, constraints handling in GA.

#### **UNIT-IV**

Evolution strategies (ES), two members non-recombinative ES, multi member ES, recombinative ES. Optimization based on swarm intelligence particle, swarm optimization and its variants.

#### **UNIT-V**

Application of soft computing techniques to problems of electrical engg. e.g. economic dispatch, reliable optimization, ANN training using evolutionary algorithms.

#### **REFERENCE BOOKS:-**

1. R. Y. Rubinstein Simulation and the Monte Carlo method, John Wiley & sons 1st Edition.
2. Paul. L. Mayer-Introducing probability and statistical application, Addison Wesley.
3. Rajasekaran and pai-Neural Network, Fuzzy logic & Genetic Algorithms. PHI Learning
4. Li Min. Fu, Neural Networks in Computer Intelligence, 9th Reprint TMH
5. Multi objective optimization using evolutionary algorithm-Kalyanmoy Deb John Wiley Sons Ltd.
6. Probability and Random processes for Electrical Engineering, Alberto Leon Garcia IInd Pearson.
7. Principles of soft computing-SN Shivanandan, SN Deepa Wiley India (P) Ltd, I edition 2007.
8. Handbook of genetic algorithm-Rajasekharans, vijayalaxmi pai.